

Omar Elmetwalli

📍 6th of October, Giza, Egypt 📞 +20 1020744196 ✉ omar.a.elmetwalli@gmail.com
🏠 omar-elmetwalli 🌐 Technical Portfolio

EDUCATION

University of Science and Technology (Zewail City) Expected Jun 2027
B.Sc. Aerospace Engineering (Top 10 Individual Rank) Giza, Egypt
GPA: 3.39/4.0
Core Coursework: Fluid Mechanics, Gas Dynamics, Heat Transfer, Numerical Analysis, ODEs/PDEs, Classical Control, Engineering Dynamics, Mechanics of Materials, Machine Design I-II.

RESEARCH COLLABORATION

Fluid Dynamics & Turbulence Research Group 2024 – Present
Collaborators: Dr. Mahmoud Ayyad (Stevens Institute), Ahmed Hamada (UT Dallas) Remote / International

- **High-Fidelity CFD:** Perform turbulence modeling and research using computational methods to investigate flow physics in [mention regime, e.g., wall-bounded/internal flows].
- **Data-Driven Fluids:** Post-process high-dimensional CFD datasets using MATLAB to extract features for Machine Learning (ML) model training and verification.
- **Physical Validation:** Verify the consistency of ML-generated results against fundamental Navier-Stokes solutions and empirical correlations to ensure thermodynamic and physical realism.
- **Numerical Solver Development:** Getting learn in OpenFOAM for custom solver implementation and advanced polyhedral meshing techniques.
- **Scholarly Communication:** Collaborate on a research manuscript targeting publication (2026) focusing on the intersection of turbulence modeling and numerical stability.

CFD & COMPUTATIONAL FLUID PROJECTS

Rocket Nozzle Flow & Structural Stress Analysis — ANSYS Fluent, MATLAB Oct – Nov 2024

- Executed density-based CFD simulations to evaluate supersonic flow field characteristics and shock wave formation in CD nozzles.
- Analyzed pressure distributions and computed wall shear stresses to verify structural integrity under extreme thermal and pressure loading.
- Conducted **mesh independence studies** to ensure numerical convergence and spatial accuracy of the results.

Optimization of Turbine Blade Cooling — ANSYS Fluent, DesignModeler Apr 2025

- Parametrically modeled internal cooling passages and conducted **Conjugate Heat Transfer (CHT)** simulations to enhance cooling effectiveness.
- Iteratively optimized geometry to maximize heat dissipation while minimizing pressure drop across the internal channels.

Cold Gas Rocket Engine: CD Nozzle & Blowdown Simulation — MATLAB Jan 2026

- Sized a Convergent-Divergent nozzle using isentropic flow relations for a 100 bar design point with perfect expansion.
- Built a time-marching rigid-tank blowdown model to predict transient mass flow rate, thrust history, and Rayleigh heat-addition losses.

Ramjet Engine: Two-Ramp Inlet Nozzle Optimization — MATLAB Jan 2026

- Optimized a fixed-geometry ramjet for $M_\infty = 2.75$ using a grid-search algorithm over inlet ramp angles to maximize total-pressure recovery.
- Predicted net thrust performance and shock interaction patterns using 1D gas dynamics and shock-expansion theory.

INDUSTRY EXPERIENCE

Anasia Group Aug – Sep 2025
Process Automation Intern (Valmet Exposure) Cairo, Egypt

- Gained hands-on exposure to **control valve technology** and instrumentation within industrial process loops.
- Designed and simulated a milk pasteurization process concept, integrating thermal control loops and automation logic.

Zewail City Mechanical Lab

Research Assistant & Design Intern

Mar 2024 – Present

Giza, Egypt

- Developed manufacturable mechanical components using Autodesk Inventor, focusing on **Design for Manufacturing (DFM)** and prototyping.

HONORS, ACTIVITIES & CERTIFICATIONS

- **Honors:** Third Place (Team), ROTORTHON 2024 — Designed VTOL Urban Air Mobility concept.
- **Leadership:** Chair & Founder of AIAA Student Branch at Zewail City (2024–Present).
- **Certification:** Introduction to Simulation (Cornell University) — FEA/CFD Foundations (2024).
- **Training:** Aircraft Maintenance Engineering Trainee at AEROSKY Academy (2024).

REFERENCES

Dr. Mahmoud Ayyad

Research Assistant Professor

Stevens Institute of Technology

Expertise: Mechanical/Fluids Engineering

mayyad@stevens.edu

Ahmed Hamada

PhD Researcher (Aerospace Engineering)

University of Texas at Dallas

Expertise: CFD & Turbulence Modeling

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Ahmed Taweel

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